P vs. NP isn't the most fundamental problem—it's the greatest obstacle

———— Reply to the comment by Allender and Williams

Ke Xu, Guangyan Zhou

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The most important contribution of our paper [5] lies in redefining the foundational problem of computer science—namely, identifying the right question to ask when a given problem is already computable. To be specific, the most fundamental problem in computer science is non-brute-force computation vs. brute-force computation, rather than P vs. NP. Our paper and current complexity theory are two paths that extend different frameworks from the same starting point in computer science. Specifically, non-brute-force computation vs. brute-force computation is an extension of Gödel's framework, while P vs. NP is an extension of Turing's framework. Although P vs. NP is a renowned Millennium Prize Problem, we aim not to emphasize it in our paper. The core argument of our paper is that P vs NP isn't the most fundamental problem—it's the greatest obstacle. We believe that any discussion is feasible, but first we must identify the right problem—for the problem itself matters more than the result.

With respect to the assumption, we have clearly elaborated on why a mathematical assumption is essential for proving computational hardness results in Appendix A of our paper as follows: *In fact, the Turing machine itself is an assumption about machines (i.e., mechanized finite formal systems) that can be realized in the physical universe. In essence, the Turing machine represents a fundamental physical assumption, and Turing's findings on uncomputability signify the physical limits of mechanically solving all instances of a problem. The computational hardness of concrete examples arises from the limits of reasoning. It is unnecessary to discuss the mechanical solving of these examples based on the concept of Turing machines. This is because if the reasoning of concrete examples requires a long time, then solving these examples mechanically will also take a long time. We can also say that computational hardness results are essentially mathematical impossibility results of reasoning. Any mathematical impossibility result cannot be proved without restrictions on the form of mathematics. To derive such results, it is essential to make a mathematical (rather than physical) assumption about reasoning.*

Three experts also addressed the issue of the assumption in their comments.

Shaowei Cai: *To establish complexity lower bounds, the authors rely on a single assumption regarding the exact algorithms for solving CSPs — "We only need to assume that this task is finished by dividing the original problem into subproblems". This should be true in algorithm design. As we know, at least until now, all exact algorithms for CSPs are based on divide-and-conquer style, such as backtracking methods.*

Zhiguo Fu: *In addition, I find the assumption about CSPs made in this paper to be reasonable and acceptable.*

Bin Wang: *I find the mathematical derivations in Xu and Zhou's paper to be correct. Additionally, this paper introduces a mathematical assumption to its reasoning, which I find both necessary and well-justified. The challenge of proving impossibility results has a long history in mathematics. For example, it took over two thousand years to prove the impossibility of trisecting an arbitrary angle using only a straightedge and compass. Such impossibility results are inherently conditional—they hold under specific mathematical frameworks rather than in an unrestricted mathematical setting. The authors* *argue that the Turing machine represents a fundamental physical assumption. This perspective seems reasonable to me, as the Turing machine, in my view, does not impose any restrictions on mathematics. Incompressibility results are a type of mathematical impossibility result, and proving them requires introducing an assumption that restricts the mathematical framework. This necessity aligns with the historical approach to proving impossibility results.*

Our main results hold under the assumption about exact algorithms for CSPs. This assumption is consistent with the reality of exact algorithms for CSPs. Allender and Williams [1] claim that there might be another sort of algorithm violating this assumption. But they also say that "We do not claim that an algorithm of this sort is likely to succeed". So our assumption has not been violated until now. For the algorithm proposed by Williams [3], it seems that this algorithm requires exponential space. Researchers do not include this type of algorithm in their discussions of exact algorithms for CSPs, as even the exhaustive algorithm for CSPs requires only polynomial space. Currently, no known algorithm can solve CSP in $O(d^{cn})$ time for some constant $c < 1$ when the domain size d is very large. This is very similar to the case of k-SAT with very large k. For details, see Ref. [2].

In [5], we constructed self-referential instances that inherently require exhaustive search, as in these cases no proper part can serve as a substitute for the whole. This approach is closely related to that of [4], which pursued complexity lower bounds through the construction of so-called "bad inputs". However, our instances go beyond merely "bad inputs"; they constitute the worst-case inputs for any computation.

The goal of our paper [5] is not to claim that we have solved a Millennium Prize Problem, then sit back and await fortune and fame. Rather, our purpose is to illuminate what constitutes the foundational problem to pursue. We hope that people, especially bright young minds, will not continue to waste their time on a non-foundational problem. Many people have extensive experience in handling numerous cases involving flawed proofs of the P vs. NP problem. The case of our paper [5], however, is entirely distinct from previous ones: The core issue lies not in whether a proof of P vs. NP is correct, but rather in whether P vs. NP itself constitutes the right question to pose. Thus, we call for broader participation and advocate for full openness and transparency in all aspects.

Finally, we believe that any discussion conducted around academic issues themselves is bound to have a positive impact on the development of academia, even though the process may be quite tortuous. Prof. Gregory Chaitin once told us: "*Of course, you are having problems convincing the community, which is only natural. It will take time, but you should persist!*" Everything that is happening now proves how correct his foresight was.

References

[1] E. Allender and R. Williams. Comment on "SAT Requires Exhaustive Search". https://people.cs.rutgers.edu/∼allender/papers/allender.williams.pdf

[2] T. Hertli, I. Hurbain, S. Millius, R. A. Moser, D. Scheder, M. Szedlák. The PPSZ algorithm for constraint satisfaction problems on more than two colors. In International Conference on Principles and Practice of Constraint Programming. Cham: Springer International Publishing, 2016: 421-437.

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